

SmartCafe

MERN Stack · Intern Group · Task Division: Weeks 4 – 7

Abishek	Chahana	Nirmal	Samikshya	Bimal	Sandesh	Nishesh	Samratt
Frontend	Frontend	Backend	Full Stack	Full Stack	Full Stack	Database	Backend
W1 Setup	W2 Backend Base	W3 Frontend Base	W4 Auth & Menu	W5 Cart & Orders	W6 Reviews & Booking	W7 QA & Deploy	W8+ Complete

Week 4

Auth, Menu UI & Core API

Completion: 40%

Abishek

Frontend Developer

[feature/abishek-week4](#)

- 1 Bootstrap React app with Vite, configure React Router v6 and Tailwind CSS
- 2 Build Login and Register pages with form validation using React Hook Form
- 3 Build Navbar component — logo, nav links, login/logout, role-aware admin link
- 4 Build ProtectedRoute and AdminRoute wrapper components with redirect logic

Chahana

Frontend Developer

[feature/chahana-week4](#)

- 1 Build MenuPage — grid of menu items fetched from API, category tab filters, search bar
- 2 Build MenuItemCard component — image, name, price, category badge, Add to Cart button
- 3 Build AdminMenuPage — table of items with Edit / Delete actions and Add New Item button
- 4 Build AddEditMenuItemForm — image preview, all fields, validation, submit to API

Nirmal

Backend Developer

[feature/nirmal-week4](#)

- 1 Set up Node.js/Express project with MVC structure, dotenv, helmet, morgan, CORS
- 2 Implement user registration & login API with bcrypt password hashing and JWT tokens
- 3 Set up auth middleware (verifyToken) and role-based access control (admin/customer)
- 4 Build Menu CRUD API — POST/GET/PUT/DELETE /api/menu with admin-only write guard

Samikshya

Full Stack Developer

[feature/samikshya-week4](#)

- 1 Create User Mongoose model — name, email, password (hashed), role, timestamps
- 2 Create MenuItem Mongoose model — name, description, price, category, imageUrl, isAvailable
- 3 Build GET /api/menu?category=&search;= with filter, search and pagination support
- 4 Set up Axios instance on frontend with JWT interceptor (auto-attach token, refresh on 401)

Bimal

Full Stack Developer

[feature/bimal-week4](#)

- 1 Build Category API — GET /api/categories; seed initial categories (Drinks, Food, Snacks, Desserts)
- 2 Build GET/PUT /api/users/profile — view and update own profile (authenticated users)
- 3 Set up Redux Toolkit store with authSlice (user, token, isAuthenticated) and cartSlice scaffold
- 4 Build AuthContext — store user/token, expose login/logout helpers, persist to localStorage

Sandesh

Full Stack Developer

[feature/sandesh-week4](#)

1	Configure Cloudinary + Multer for menu item image upload; integrate with create/update API
2	Write express-validator input validation middleware for all auth and menu routes
3	Set up global error handling middleware and standard API response format {success, data, message}
4	Build admin-only GET /api/users — list all users with pagination and role filter

Nishesh		Database Engineer	feature/nishesh-week4
1	Design MongoDB collections: users, menuitems, categories — fields, indexes, relationships diagram		
2	Write Mongoose seed script — 5 categories and 20 sample menu items with placeholder images		
3	Add indexes: menuitems.category, menuitems.isAvailable, users.email (unique)		
4	Update project README with DB schema diagram, collection descriptions and local setup guide		

Samratt		Backend Developer	feature/samratt-week4
1	Configure Jest + Supertest; write integration tests for register, login and profile endpoints		
2	Write Jest tests for menu CRUD — create, read, update, delete and category filter		
3	Set up express-async-errors for global uncaught async exception handling		
4	Build admin-only PATCH /api/admin/users/:id/role — promote or demote user role		

End of Week 4 — Must Be Complete

■	Login & Register pages working end-to-end
■	Menu page renders with live data from MongoDB
■	Admin can add / edit / delete menu items
■	JWT auth with protected routes enforced
■	Category filter working on frontend
■	All auth and menu API tests passing

Week 5		Cart, Orders & Payment	Completion: 60%
Abishek		Frontend Developer	feature/abishek-week5
1	Build CartPage — item list with quantity stepper, remove button, subtotal per item and cart total		
2	Build CartSummary sidebar — item count, subtotal, proceed to checkout button		
3	Wire cart actions to Redux cartSlice — addItem, updateQuantity, removeItem, clearCart		
4	Build toast notification system for cart actions (added, removed, out of stock)		
Chahana		Frontend Developer	feature/chahana-week5
1	Build CheckoutPage — address form, order summary panel, payment method selector		
2	Build PaymentMethodSelector component — radio card style with logo for eSewa, Khalti, COD		
3	Build OrderSuccessPage — order reference, summary table, estimated time, link to /orders		
4	Handle eSewa/Khalti redirect callback on frontend; show success or failure state		
Nirmal		Backend Developer	feature/nirmal-week5
1	Build Cart API — POST /api/cart/add, PATCH /api/cart/update, DELETE /api/cart/remove, GET /api/cart		
2	Enforce stock check in cart service; throw error if quantity exceeds available stock		
3	Build POST /api/orders — create order from cart, reduce item stock, clear cart on success		
4	Build GET /api/orders (admin) — all orders with status filter, date range and pagination		
Samikshya		Full Stack Developer	feature/samikshya-week5
1	Create Order Mongoose model — user, items[], totalAmount, status, paymentMethod, paymentStatus		
2	Build GET /api/orders/my — current user's order history with pagination		
3	Build GET /api/orders/:id — single order detail (owner or admin only)		
4	Build PATCH /api/orders/:id/status (admin) — update status (pending/preparing/ready/delivered)		
Bimal		Full Stack Developer	feature/bimal-week5
1	Build OrderHistoryPage — list of user's past orders with status badge and date		
2	Build OrderDetailPage — full order breakdown, items, payment info and current status		
3	Listen to Socket.io orderStatusChanged event; update order status in real time on UI		
4	Sync cart with backend on login — merge local Redux cart with server cart		
Sandesh		Full Stack Developer	feature/sandesh-week5
1	Integrate eSewa payment gateway — initiate payment, verify callback, update paymentStatus		
2	Integrate Khalti payment gateway — initiate, verify webhook, update order paymentStatus		
3	Set up Socket.io server; emit orderStatusChanged event when admin updates order status		
4	Build email notification service (Nodemailer) — send order confirmation email on placement		
Nishesh		Database Engineer	feature/nishesh-week5
1	Add orders and payments collections to MongoDB; document schema and relationships in README		

2	Add indexes: orders.user, orders.status, orders.createdAt for efficient admin queries
3	Write seed script for 10 sample orders in various statuses for testing
4	Monitor query performance using MongoDB explain(); fix any missing or inefficient indexes

Samratt	Backend Developer	feature/samratt-week5
1	Write Jest integration tests for cart add, update, remove and stock guard scenarios	
2	Write Jest tests for order creation, status update and payment callback routes	
3	Send order status update email when admin changes status to ready or delivered	
4	Set up express-rate-limit on payment and order routes to prevent abuse	

End of Week 5 — Must Be Complete

■	Cart add / update / remove working end-to-end
■	Order placement API complete
■	eSewa / Khalti payment integrated (sandbox)
■	Order history visible per user
■	Admin can view and update all orders
■	Real-time order status via Socket.io
■	All cart and order tests passing

Week 6		Reviews, Table Booking & Admin Dashboard	Completion: 80%
Abishek		Frontend Developer	feature/abishek-week6
1	Build ReviewSection on MenuItemDetailPage — star rating input, comment box, submit button		
2	Display list of reviews — user name, date, star rating, comment for each review		
3	Show average rating badge and total review count on MenuItemCard and detail page		
4	Prevent duplicate review: hide review form if current user already submitted a review		
Chahana		Frontend Developer	feature/chahana-week6
1	Build TableBookingPage — date picker, time slot selector, guest count, available tables grid		
2	Build MyBookingsPage — list of user's bookings with status badge and cancel option		
3	Build AdminBookingsPage — all bookings in a table with confirm/cancel action buttons		
4	Show booking confirmation toast and redirect user to /my-bookings after successful booking		
Nirmal		Backend Developer	feature/nirmal-week6
1	Create Review model — user, menuItem, rating (1-5), comment; enforce one review per user per item		
2	Build POST /api/reviews, GET /api/reviews/:menuItemid, DELETE /api/reviews/:id (admin/owner)		
3	Aggregate average rating on MenuItem after each review create/delete using Mongoose post-hooks		
4	Build admin stats API — GET /api/admin/stats: total orders, revenue, active users, popular items		
Samikshya		Full Stack Developer	feature/samikshya-week6
1	Create TableBooking model — user, tableNumber, date, timeSlot, guestCount, status, timestamps		
2	Build POST /api/bookings — create booking with conflict check (same table + date + slot)		
3	Build GET /api/bookings/my and GET /api/bookings (admin) — with pagination and status filter		
4	Build PATCH /api/bookings/:id/status (admin) — confirm or cancel booking		
Bimal		Full Stack Developer	feature/bimal-week6
1	Build Admin Dashboard page — stat cards (orders, revenue, users, bookings) fetched from API		
2	Integrate Recharts — line chart for daily revenue, bar chart for top 5 selling items		
3	Build RecentOrdersTable widget on dashboard — last 10 orders with status and amount		
4	Build AdminSidebar with navigation links to Dashboard, Menu, Orders, Bookings, Users		
Sandesh		Full Stack Developer	feature/sandesh-week6
1	Build revenue chart API — GET /api/admin/revenue?period=week month returning daily totals		
2	Build GET /api/menu/:id — single menu item detail with populated reviews and average rating		
3	Build GET /api/admin/users/:id — admin view of individual user profile and order history		
4	Send booking confirmation and cancellation emails via Nodemailer		
Nishesh		Database Engineer	feature/nishesh-week6
1	Add reviews and tableBookings collections; document schema and indexes in README		

2	Add compound index on reviews (user + menuItem) to enforce one-review-per-user constraint
3	Add compound index on tableBookings (tableNumber + date + timeSlot) for conflict detection
4	Write seed data — 30 sample reviews across menu items and 10 bookings in various statuses

Samratt	Backend Developer	feature/samratt-week6
1	Write Jest tests for review create, duplicate guard, rating aggregation and delete	
2	Write Jest tests for booking conflict check, admin stats and revenue aggregation	
3	Write Jest integration tests for admin user management and menu detail with reviews	
4	Audit all API routes for missing input validation; add express-validator rules where missing	

End of Week 6 — Must Be Complete

■	Review submission and listing working per menu item
■	Average rating calculated and displayed
■	Table booking API complete with conflict check
■	Admin can confirm and cancel bookings
■	Admin dashboard shows live stats
■	Revenue chart renders with real data
■	All review and booking tests passing

Week 7		QA, Performance & Deployment	Completion: 99%
Abishek		Frontend Developer	feature/abishek-week7
1	Audit all pages for mobile responsiveness; fix layout on 375px, 768px and 1280px breakpoints		
2	Run Lighthouse on MenuPage, CheckoutPage and Dashboard; fix issues to reach 90+ score		
3	Add React lazy + Suspense code-splitting on AdminDashboard, OrderHistory and BookingPage		
4	Add loading skeleton (shimmer) components on MenuPage and OrderHistoryPage while data fetches		
Chahana		Frontend Developer	feature/chahana-week7
1	Deploy React app to Vercel; configure production API base URL via environment variable		
2	Build 404 Not Found page and global React Error Boundary component		
3	Implement optimistic UI updates for cart add/remove — instant feedback before API confirms		
4	Final UI polish — consistent spacing, font sizes, button states and colors across all pages		
Nirmal		Backend Developer	feature/nirmal-week7
1	Deploy Node.js/Express backend to Railway; configure production env vars and MongoDB Atlas URI		
2	Set up CI/CD pipeline with GitHub Actions — run tests on PR, auto-deploy on merge to main		
3	Configure CORS for production Vercel domain; smoke-test all API endpoints against deployment		
4	Set up Winston logger — log errors and slow requests (>500ms) to file in production		
Samikshya		Full Stack Developer	feature/samikshya-week7
1	Write Cypress e2e test — full user flow: register → browse menu → add to cart → checkout → success		
2	Write Cypress e2e test — admin flow: login → add menu item → view orders → update status		
3	Write Cypress e2e test — review flow: login → open item → submit review → see updated rating		
4	Fix all bugs found during Cypress testing; re-run suite until all tests are green		
Bimal		Full Stack Developer	feature/bimal-week7
1	Write Cypress e2e test — booking flow: select date/time → confirm → view in my bookings		
2	Implement infinite scroll or pagination on MenuPage and OrderHistoryPage		
3	Add React Helmet meta tags for SEO on Home, Menu and item detail pages		
4	Final cross-browser testing (Chrome, Firefox, Safari); fix any visual inconsistencies		
Sandesh		Full Stack Developer	feature/sandesh-week7
1	Audit all API endpoints for missing auth guards and security gaps; fix all issues found		
2	Run load test with Artillery on /api/menu, /api/orders, /api/cart — ensure <300ms at 50 concurrent		
3	Add helmet, hpp and tightened rate-limit config for production; verify no sensitive data in responses		
4	Write final API documentation — Postman collection covering all endpoints with example responses		
Nishesh		Database Engineer	feature/nishesh-week7
1	Migrate local MongoDB data to MongoDB Atlas; set up production cluster with automated backups		

2	Configure Atlas IP whitelist for Railway/Render backend; test connection from deployed server
3	Run MongoDB Atlas Performance Advisor; create recommended indexes and document all changes
4	Write final DB README — schema diagrams, index list, seed instructions and Atlas setup guide

Samratt		Backend Developer	feature/samratt-week7
1	Run full Jest test suite; ensure all existing unit and integration tests pass before deployment		
2	Add missing Jest tests to reach minimum 80% code coverage across routes and services		
3	Document environment variable setup, deployment steps and production checklist in README		
4	Final code review across all branches — remove console.logs, hardcoded secrets, unused imports		

End of Week 7 — Must Be Complete

■	All pages fully responsive (mobile, tablet, desktop)
■	Cypress e2e tests passing for all critical flows
■	Lighthouse score 90+ on Performance and Accessibility
■	Backend deployed to Railway / Render
■	Frontend deployed to Vercel
■	MongoDB Atlas production database configured
■	API response times under 300ms for core endpoints
■	Final README and API documentation complete